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Scaling law for penetration depth in laser welding

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Outline

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Highlights

- Proposed a scaling law for penetration depth in laser welding.
- Accounted for multiple reflections phenomena.
- Validated the law using extensive and systematic experiments.

Abstract

A simple scaling law for penetration depth in laser welding is proposed considering heat flow characteristics and multiple reflections. First, a process parameter is identified that is proportional to the surface temperature during laser processing, and the parameter is modified by accounting for the effect of multiple reflections. As a result, the normalized

penetration depth is expressed as a function of a single parameter that is a combination of laser intensity, interaction time and an indicator of the strength of multiple reflections. The obtained scaling law is applicable not only to conduction mode welding but also to keyhole mode welding, and provides insight into why and how penetration depth changes in a particular way. Systematic and extensive welding experiments were conducted using a 2kW multi-mode fiber laser and two types of steels. The experimental results were in good agreement with the proposed scaling law.

Keywords

Laser welding; Scaling law; Penetration depth; Multiple reflections

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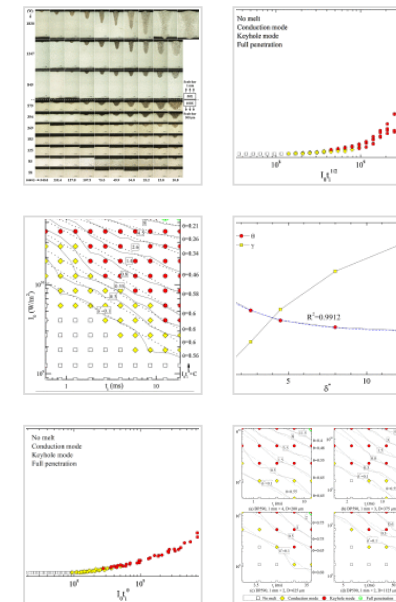
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1. Introduction

Prediction of penetration depth is arguably one of the most important tasks in the study of laser welding because it is the first thing that needs to be determined in most welding applications. Unfortunately, however, laser welding is a very complex process (where melt flow, evaporating vapor flow, heat transfer, phase change, and laser beam interaction with plasmas, and multiple reflections of laser beam occur simultaneously), so this task in most cases is very challenging. Although there are many sophisticated simulation models, these models are generally too costly to be used for this purpose and simple predictive models are much preferred.

[Benyounis et al. \(2005\)](#) obtained linear and quadratic equations for predicting heat input and weld-bead geometry by studying the effect of laser power, welding speed and focal position using the response surface methodology. [Hann et al. \(2011\)](#) presented a simple physical model to predict melt depth and width by using the concept of mean surface

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enthalpy. [Assuncao et al. \(2012\)](#) studied the transition from conduction to keyhole mode welding by investigating the change in penetration depth as power density or interaction time varies. They found that a single threshold power density does not exist and the transition occurs over a wide range of laser power densities. Recently, [Suder and Williams \(2014\)](#) proposed a simple empirical model for predicting penetration depth.

Although these models are useful, none of these models accounted for the complex energy absorption phenomena caused by a keyhole. [Kaplan \(2012\)](#) reported that a wavy keyhole surface would rapidly change the direct beam absorption from angle-dependent Fresnel absorption to its angle-averaged value. [Volpp and Vollertsen \(2013\)](#) reported by using their analytical multiple reflections model that different spatial laser intensity distributions lead to similar keyhole shapes due to similar energy distributions on the keyhole walls after multiple reflections but to different radial pressure gradients. As well known, multiple reflections phenomenon greatly affects laser beam absorption and contributes to deeper keyhole penetration.

In this study, a simple scaling law for penetration depth in laser welding is proposed, where penetration depth is expressed as a function of a single parameter. The proposed scaling law was formulated based on a simple one-dimensional heat conduction model, and the effect of multiple reflections was accounted for. Because the scaling law was obtained from a laser heating problem, its physical meaning and why it needs to be formulated that way can be clearly explained. Systematic and extensive welding experiments were conducted using a 2kW multi-mode fiber laser with a top-heat beam profile. Two types of steels and four different laser beam diameters were tested, and the obtained results were found to be in good agreement with the proposed scaling law.

2. Proposal of scaling law

Laser welding is a three-dimensional moving heat source problem by nature. To simplify the process, the process will be modeled as a simple one-dimensional heat conduction problem, where the specimen is assumed to be semi-infinite and be heated by a constant

heat flux energy source with a power density (or laser intensity) of I_0 for a time duration equal to the interaction time (t_i). If the laser beam has a top-hat profile with a beam diameter of D , and if the laser beam moves at a scanning speed of V_s , the interaction time is D/V_s . Neglecting the energy loss at the substrate surface due to convection and radiation, an analytical solution that is valid during the heating period (i.e., $0 \leq t \leq t_i$) can be obtained as follows (Incropera and DeWitt, 1990):

$$T(z, t) = T_0 + \frac{2\eta I_0 \sqrt{\alpha t}}{\sqrt{\pi k}} \exp\left(-\frac{z^2}{4\alpha t}\right) - \frac{\eta I_0 z}{k} \operatorname{erfc}\left(\frac{z}{2\sqrt{\alpha t}}\right) \quad (1)$$

Here, T , t , and z are temperature, time, and the spatial coordinate in the target's thickness direction, respectively, α and k are thermal diffusivity and thermal conductivity of the target material, respectively, T_0 is initial temperature, I_0 is laser intensity, and η is laser beam absorptivity. Note that this model is especially reasonable for conduction mode welding where the melt pool surface remains pretty much flat. Therefore, conduction mode welding is assumed first and the obtained result will be extended later to keyhole welding.

With this temperature profile, the surface temperature history can be evaluated as:

$$T_s(t) = T(z = 0, t) = T_0 + \frac{2\eta I_0 \sqrt{\alpha t}}{\sqrt{\pi k}} \quad (2)$$

Then, the increase in surface temperature at the end of the laser heating is calculated as

$$\Delta T_s = T_s(t_i) - T_0 = \frac{2\eta I_0 \sqrt{\alpha t_i}}{\sqrt{\pi k}} \quad (3)$$

Neglecting material properties and constants, the following relation can be obtained:

$$\Delta T_s \sim \eta I_0 t_i^{1/2} \quad (4)$$

Note that laser intensity and interaction time are the two primary parameters for any laser material processing, and the amount of temperature increase seems to be closely associated with η and $I_0 t_i^{1/2}$. Although the specimen is assumed semi-infinite, this result

is valid for most welding problems because the heat penetration depth during the interaction time is small compared to the specimen thickness.

In laser welding, the amount of surface temperature rise determines the type of physical phenomena, such as melting and vaporization. Also, the penetration depth strongly depends on this quantity. In a one-dimensional heat conduction problem, once the temperature profile at the end of the heating period is known, it is as though the penetration depth is also determined. Therefore, we propose the following functional relationship for penetration depth:

$$\delta = \hat{f}(\eta I_0 t_i^{1/2}) \quad (5)$$

In obtaining Eq. (5), one important fact has been ignored regarding the one-dimensional heat flow model: Even when the surface temperatures are the same, the melt penetration depth is shorter if the beam size is smaller, because in this case a lot of heat flows laterally rather than in the specimen thickness direction. Note that the maximum melt penetration depth is achieved when one-dimensional heat flow can be assumed (i.e., the beam is very large compared to the problem size), but if the beam is very small, the laser beam is considered a point heat source and heat flows in all directions uniformly and the melt penetration depth decreases. Therefore, assuming the penetration depth is proportional to the beam diameter, Eq. (5) is modified as

$$\delta^* \equiv \frac{\delta}{D} = \tilde{f}(\eta I_0 t_i^{1/2}), \quad (6)$$

where δ^* is the normalized penetration depth.

So far, only the conduction mode welding has been considered. As the surface temperature increases beyond the material's boiling temperature, a keyhole starts to form and the laser beam undergoes multiple reflections inside the keyhole. [Ki et al. \(2002b\)](#) studied how multiple reflections phenomena change the laser beam absorption characteristic using their self-consistent welding model. They found that as the keyhole deepens the effective laser absorptivity (ELA) increases and the normalized maximum

intensity (NMI) (which is the maximum intensity on the keyhole surface that is normalized by the laser beam's original centerline intensity I_0) also increases. Based on their findings, it can be assumed that the laser intensity (based on the increased NMI) and the laser absorptivity (based on the increased ELA) both increase as the keyhole deepens. Since I_0 is used as the original laser beam intensity, the following substitution is proposed to account for the effect of multiple reflections:

$$\eta I_0 \rightarrow I_0^{\gamma+1} \quad (7)$$

Here, the net effect of multiple reflections (due to both increased absorptivity and increased intensity) is modeled as I_0^γ , where γ denotes the strength of multiple reflections and is believed to increase from 0 as the penetration depth increases ($\gamma \geq 0$). This relation is useful because the effective laser absorptivity η is difficult to estimate when a keyhole is formed, so it needs to be expressed in terms of other easily calculable process parameters. From Eqs. (6), (7), the following scaling law for penetration depth is obtained:

$$\delta^* = \tilde{f}(I_0^{\gamma+1} t_i^{1/2}), \quad (8)$$

where $\gamma=0$ for conduction mode welding and $\gamma>0$ when a keyhole is formed. Here, Eq. (8) can be modified as follows:

$$\delta^* = \tilde{f}(I_0^{\gamma+1} t_i^{1/2}) = \tilde{f}([I_0 t_i^{1/(2(\gamma+1))}]^{\gamma+1}) = f(I_0 t_i^{1/(2(\gamma+1))}), \quad (9)$$

where

$$f(x) = \tilde{f}(x^{\gamma+1}). \quad (10)$$

By defining θ as

$$\theta \equiv \frac{1}{2(\gamma+1)}, \quad (11)$$

Eq. (9) is re-written as

$$\delta^* = f(I_0 t_i^\theta). \quad (12)$$

In this study, $I_0 t_i^\theta$ will be used as the primary parameter for predicting penetration depth. Note that, unlike γ , θ decreases from 0.5 as the effect of multiple reflections becomes stronger.

3. Experimental tests and discussion

To test the proposed scaling law, laser welding experiments were systematically conducted using a 2kW multi-mode fiber laser (IPG YLS 2000) with a wavelength of 1070nm and a top-hat focused beam profile. The beam was focused on the top surface of specimens, and the optical setups used are summarized in [Table 1](#).

Table 1. Optical setups used for experiments.

Fiber diameter (μm)	Focal length (mm)	Collimation length (mm)	Focused beam diameter (μm)
200	160	160	200
200	300	160	375
200	500	160	625
600	300	160	1125

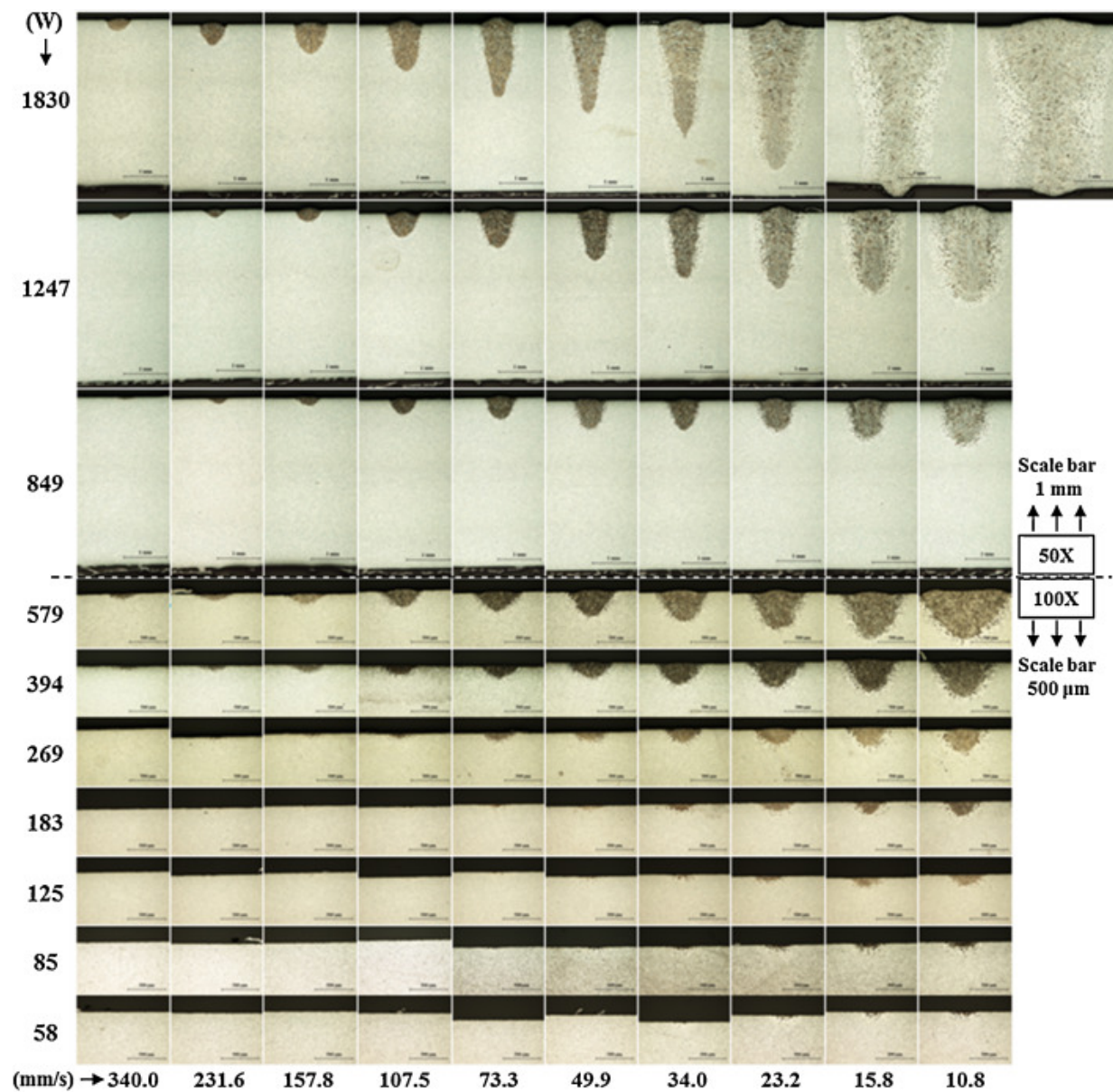
As the first test, bead-on-plate welding was conducted on 4mm thick PO steel plates, and a focused beam diameter of 200 μm was used to achieve deep penetration. The chemical composition of PO steel is summarized in [Table 2](#). In this study, a systematic experiment was designed, where laser intensity from 27W ($I_0=8.6\times 10^4\text{W}/\text{cm}^2$) to 1830W ($I_0=5.8\times 10^6\text{W}/\text{cm}^2$) and beam scanning speed from 10.8mm/s ($t_i=18.5\text{ms}$) to 340mm/s

($t_i=0.588\text{ms}$) were respectively divided into 12 and 10 equally spaced points on a logarithmic scale, so a total of 120 experiments were carried out. These 120 experiments encompasses almost the full welding capacity of the laser. To avoid the shield gas effect, no shield gas was employed.

Table 2. Chemical composition (%) of PO steel.

C	Si	Mn	P	S	Fe
0.0304	0.004	0.225	0.007	0.004	Balance

[Fig. 1](#) presents the optical micrographs of cross-sections of welded specimens arranged using laser power and scanning speed as vertical and horizontal axes respectively. Because melting was not observed for the lowest two powers, these two cases are omitted and only 100 images are presented in [Fig. 1](#). In this figure, along the diagonal lines (when drawn from top left to bottom right) energy density values are designed to be the same, and the melt penetration depth is shown to increase as the energy density increases.



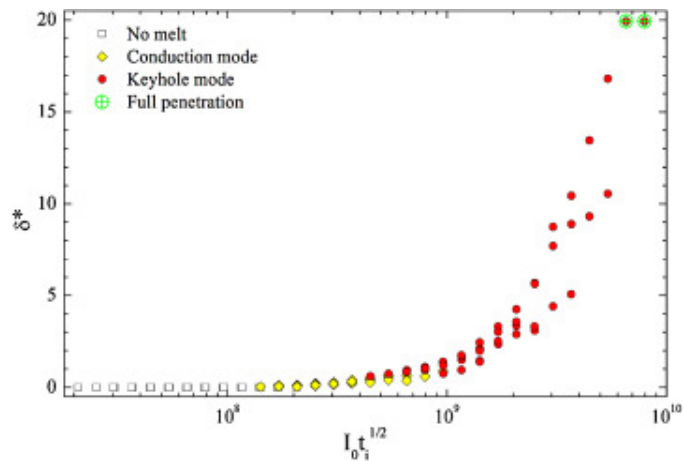
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Fig. 1. Optical micrographs of the cross-sections of 4mm thick PO steel plates. Images are arranged using laser power as the vertical axis and the scanning speed as the horizontal

axis.

Fig. 2 presents the normalized penetration depth δ^* versus $I_0 t_i^{1/2}$, where penetration depths were measured by observing the optical micrographs. Considering that these experimental results cover a very large process parameter space, this figure shows a good correlation although the correlation becomes more and more inaccurate as the penetration depth increases. It is believed that this mismatch is caused by the increased role of multiple reflections as the surface temperature rises.



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Fig. 2. Normalized penetration depth δ^* vs. $I_0 t_i^{1/2}$ obtained for 4mm thick PO steel plates.

In order to investigate the effect of multiple reflections, the normalized penetration depth δ^* is now presented as contour lines on an I_0-t_i diagram in Fig. 3. In this figure, all 120 individual experiments are expressed as colored circles, where white, yellow, red and green colors respectively represent no melting, conduction mode welding, keyhole mode welding and full penetration. The arrangement of the data points are the same as the arrangement of the images in Fig. 1, and the type of welding was visually investigated by using the high-speed photography (Ki et al., 2002a) during the experiment. In Fig. 3, black

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To investigate how the contour line slope changes on the diagram, in this study each contour line is fitted by a linear line (shown as dashed lines) that has roughly the same average slope as the contour line. Once a fitting line is obtained for each contour line, the equation for the line can be calculated. Because a log–log plot is used for Fig. 3, the following is the equation for the line:

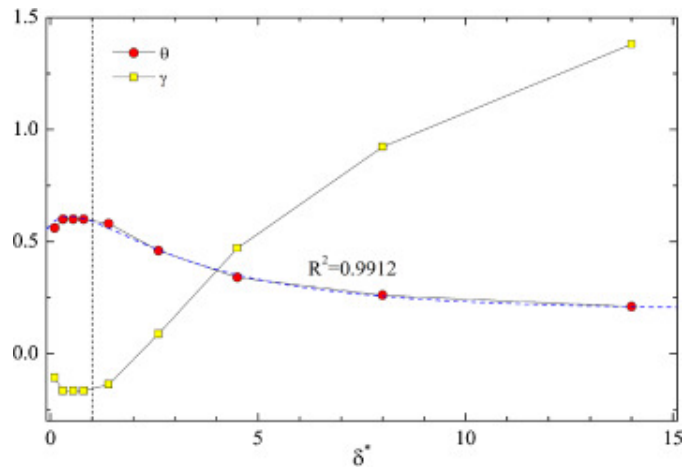
$$\log(I_0) = -\theta \log(t_i) + b, \quad (13)$$

where θ is the absolute value of the slope and b is the I_0 -intercept. Rearranging,

$$I_0 t_i^\theta = 10^b = \text{const}, \quad (14)$$

which is an approximated equation from which the given normalized penetration depth can be obtained. As suggested in Section 2, Eq. (14) shows that the normalized penetration depth can be expressed primarily by one parameter that is a combination of laser intensity and interaction time, but the parameter includes a variable (θ) that is a function of normalized penetration depth.

On the right side of Fig. 3, θ values for the given set of contour lines are presented, which are plotted versus δ^* in Fig. 4. As shown, θ increases slightly from 0.56 to 0.6 and then decreases all the way to 0.21. Note that, comparing with Fig. 3, the first three points (θ from 0.56 to 0.6) correspond to the conduction mode welding, the fourth point the transition region ($\theta=0.6$), and the rest belongs to the keyhole mode welding (θ from 0.58 to 0.21). It was predicted in the previous section that $\theta=0.5$ when a keyhole is not formed, and the experimental result showed a slightly higher value and it is believed that the mismatch is due to both experimental errors and the limitations of the one-dimensional heat flow model.



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Fig. 4. θ and γ vs. normalized penetration depth δ^* obtained from Fig. 3.

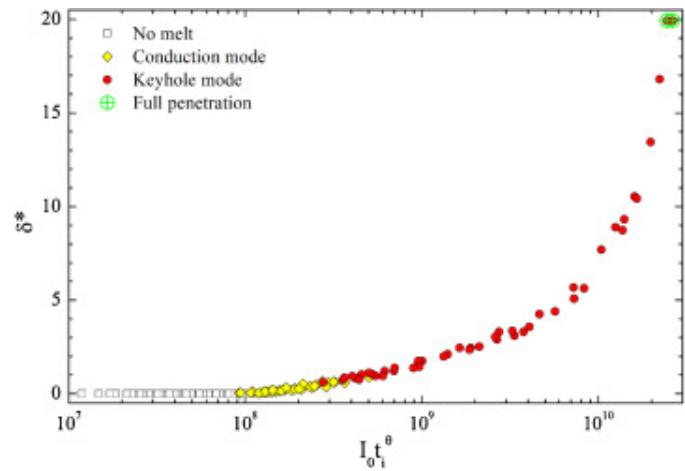
Fig. 4 also presents γ values as a function of δ^* . Recall that γ is an indicator of how strong the effect of multiple reflections on penetration depth is. As shown in the figure, multiple reflections become more dominant as the penetration depth increases, and it increases up to 1.38 when δ^* is 14, meaning that the influence of multiple reflections can be represented by $I_0^{1.38}$. Note also that γ is negative when δ^* is small as discussed earlier.

In this study, θ values in Fig. 4 were fitted by using the following function,

$$\theta(\delta^*) = 0.1940 + 4.280(\delta^* + 0.5972)^{1.2} \exp[-2.328(\delta^* + 0.5972)^{0.5}], \quad (15)$$

which is shown as a blue dashed line. Using this, δ^* is plotted versus $I_0 t_i^\theta$ in Fig. 5.

Compared with Fig. 2, this figure shows a much improved correlation. Unlike Fig. 2, the obtained result is very accurate even in the keyhole mode region.



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Fig. 5. Normalized penetration depth δ^* vs. $I_0 t_i^9$ obtained for 4mm thick PO steel plates.

In order to investigate the effect of beam diameter, a similar experiment was conducted using four different beam diameters as summarized in Table 3. In this case, instead of using a thick plate, several 1 mm thick DP 590 dual phase steel sheets were stacked. The chemical composition of the steel is provided in Table 4. For each beam diameter, a total of 25 experiments were conducted using five laser powers and five scanning speeds. Depending on the laser beam power, up to four sheets were stacked to obtain as deep penetration as possible. In this study, no shield gas was employed in order to avoid the shield gas effect.

Table 3. Experimental parameters for the laser welding of stacked 1 mm thick DP590 steel sheets.

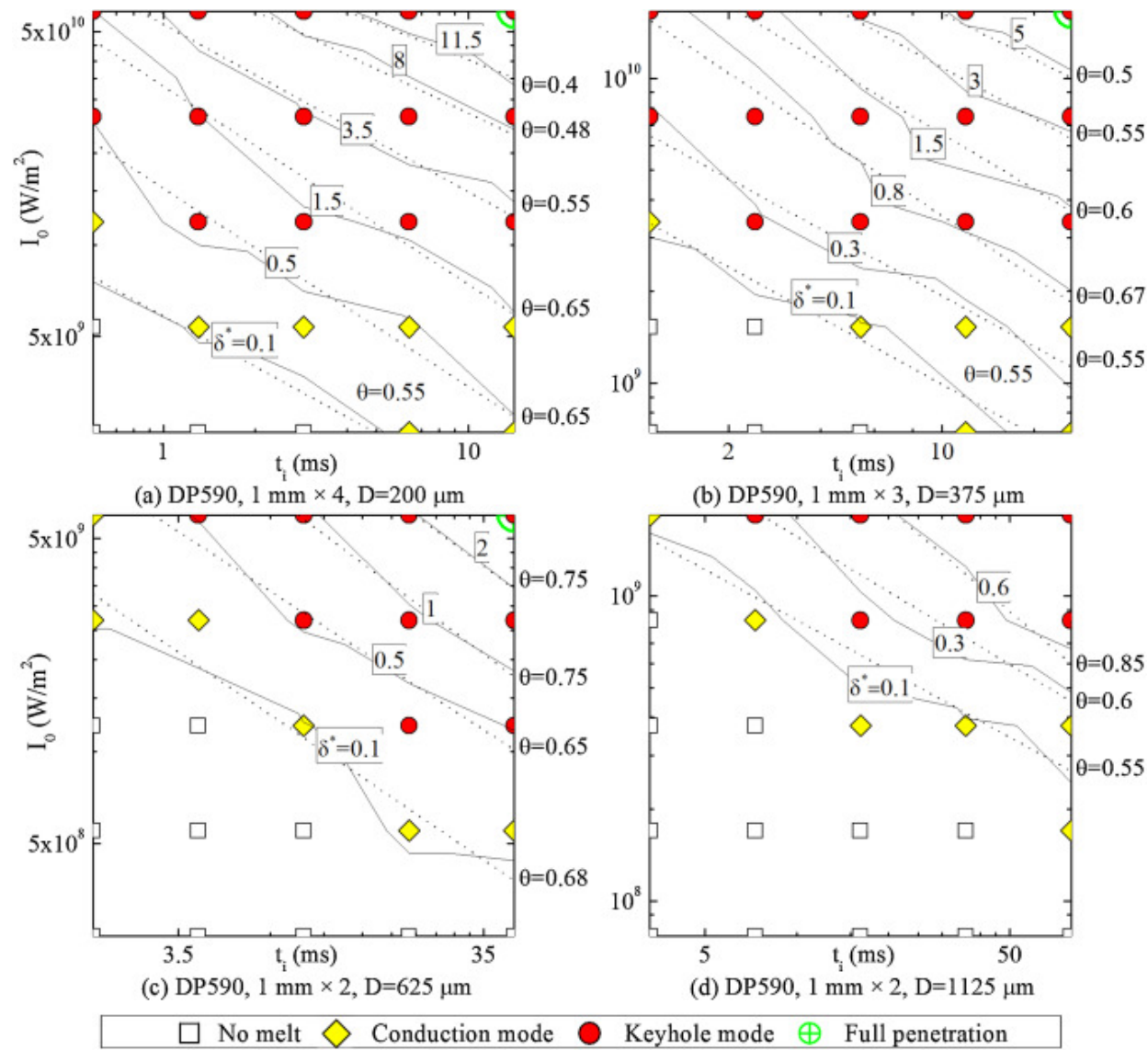
Beam diameter (μm)	Laser power (W)	Scanning speed (mm/s)	Steel type	Thickness
200	76, 169, 374, 827, 1830	14.2, 31.4, 69.4, 153.6, 340.0	DP 590	1 mm×4

Beam diameter (μm)	Laser power (W)	Scanning speed (mm/s)	Steel type	Thickness
375				1 mm $\times$ 3
625				1 mm $\times$ 2
1125				1 mm $\times$ 2

Table 4. Chemical composition (%) of DP 590 steel.

C	Si	Mn	P	S	Fe
0.078	0.0345	1.796	0.0128	0.0014	Balance

Just like the PO steel experiment, the penetration depth was measured for each experiment, and the normalized penetration depth δ^* is presented as contour lines on an I_0-t_i diagram in Fig. 6. In this figure, welding types are expressed as colored circles following the same color convention used so far, and it can be seen that a total of 100 experiments were conducted over a large process parameter space within the capacity of the 2 kW fiber laser system. Each contour line was approximated by a linear line and its slope θ was calculated using Eq. (13).



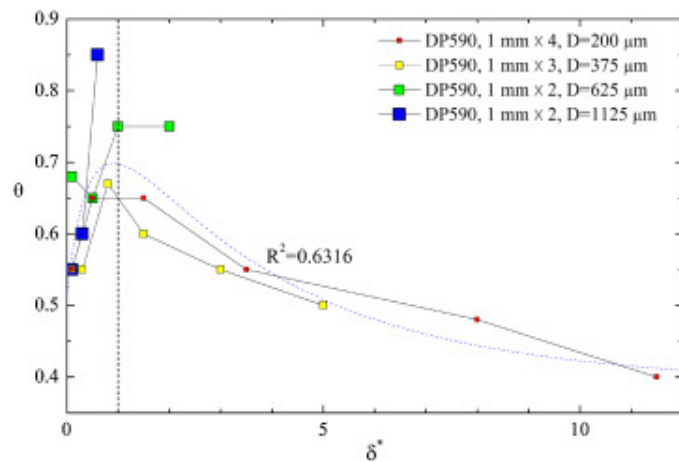
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Fig. 6. Contour lines of normalized penetration depth δ^* (solid lines) and their linear approximations (dashed lines) for stacked DP 590 steel sheets.

Fig. 7 presents θ versus δ^* for the four different laser beam diameters. As shown in the figure, it can be noticed that the attainable penetration depth decreases as the beam diameter increases. For the smallest two beam diameter results, θ increases slightly and then decreases monotonically as δ^* increases. For 625 and 1125 μm diameters, the laser power was not high enough to obtain deep penetration, but θ shows an increasing pattern when $\delta^* \leq 1$. For all four cases, it can be noticed that θ increases and reaches the maximum value at $\delta^* \approx 1$ and then decreases beyond that point. In Fig. 7, the blue dashed line shows a fitting curve, which has the following form:

$$\theta(\delta^*) = 0.3847 + 12.198(\delta^* + 0.1473)^{1.5} \exp[-3.662(\delta^* + 0.1473)^{0.4}] \quad (16)$$



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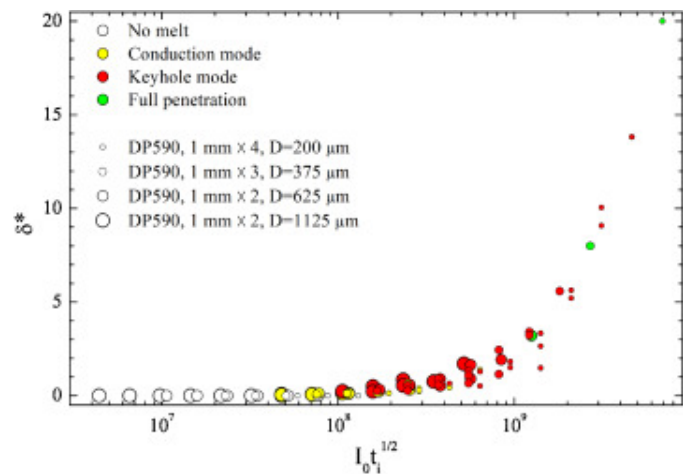
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Fig. 7. θ vs. normalized penetration depth δ^* for stacked DP 590 steel sheets obtained from Fig. 6.

Note that the overall pattern of $\theta(\delta^*)$ is qualitatively similar to the PO steel result, and Eq. (16) has a similar functional form to Eq. (15). Considering that stacked steel sheets were laser-welded instead of a thick steel plate, the obtained result seems reasonable. In this case, it is clearly observed that $\theta(\delta^*)$ peaks at near $\delta^* \approx 1$, especially when the beam

diameter is large. In fact, a small peak was also observed near $\delta^* \approx 1$ for the PO steel result as shown in Fig. 4 ($\delta^* = 1$ is shown as a vertical dashed line in Fig. 4, Fig. 7). We believe that this is ascribed to the decreased average laser intensity when the keyhole is shallow. Note that $\delta^* \approx 1$ means that the beam diameter and the penetration depth are roughly the same. Because the beam diameter can be assumed to be the same as the keyhole diameter, especially when the keyhole is shallow, it can be said that the maximum θ occurs when the keyhole depth is roughly the same as the keyhole diameter. If the keyhole depth is smaller than a certain threshold value, multiple reflections phenomenon does not take effect, and only the surface area is increased, leading to a decreased average intensity. It is believed that this explains why γ decreases and become negative when δ^* is small compared to the beam diameter ($\delta^* \approx 1$).

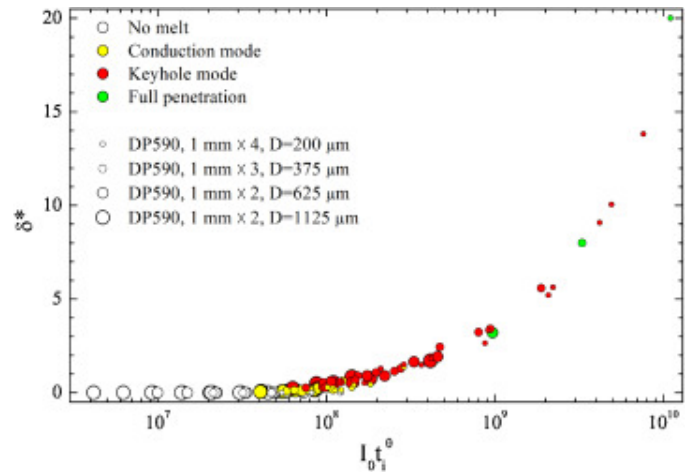
Fig. 8, Fig. 9 present δ^* versus $I_0 t_i^{1/2}$ and $I_0 t_i^\theta$, respectively, where θ is evaluated using Eq. (16). The size of a circle denotes the beam diameter. As clearly demonstrated, considering multiple reflections by modifying θ improves the correlation, especially for the keyhole mode welding region, and the proposed scaling law is found to be valid regardless of the beam diameter.



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Fig. 8. Normalized penetration depth δ^* vs. $I_0 t_i^{1/2}$ obtained for stacked DP 590 steel sheets.



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Fig. 9. Normalized penetration depth δ^* vs. $I_0 t_i^\theta$ obtained for stacked DP 590 steel sheets.

4. Conclusions

The major findings are summarized as follows:

- It has been revealed that $I_0 t_i^{1/2}$ is an important parameter in laser processing and is closely related to the surface temperature.
- The parameter $I_0 t_i^{1/2}$ can be effectively used for predicting penetration depth in the conduction mode welding regime but becomes inaccurate as the penetration depth increases and the influence of multiple reflections becomes stronger.

- For both conduction mode and keyhole mode welding, the normalized penetration depth δ^* can be expressed as a function of a single parameter $I_0 t_i^\theta$, i.e., $\delta^* = f(I_0 t_i^\theta)$, where θ is a parameter that takes into account multiple reflections.
- Theoretically, the exponent θ is 0.5 for conduction mode welding, but the experimentally obtained value were slightly higher.
- The parameter γ denotes the strength of multiple reflections, and increases as the keyhole deepens. For shallow keyholes, γ could assume negative values.

Acknowledgments

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...It can be noted that both parameters were linearly highly correlated. Kim and Ki [30] modeled the laser beam intensity ($I_{0\gamma}$, where γ is the strength of the multiple reflections) to consider the effect of multiple reflections inside the keyhole and recommended the use of $I_{0\gamma}\theta$ to predict the penetration depth. Consequently, $I_{0\gamma}^{1/3}$ was considered to be an effective parameter in understanding the laser absorptance considering multiple reflections in full-penetration laser keyhole welding....

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Oh S., ..., Ki H.

Optics and Laser Technology • Volume 113 • 2019

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Research article

Correlation of high power laser welding parameters with real weld geometry and microstructure

Liu S., ..., Jiang P.

Optics and Laser Technology • Volume 94 • 2017

Citation Excerpt:

...Welds without defects such as porosity, humping or underfilling could be produced at wide process windows [11,12]. Empirical data-based statistical analyses, model-based analytic methods and finite element simulation have also been utilized to predict the weld depth and width from different aspects [13–15]. However, weld geometry varies throughout the whole thickness....

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Research article

Selective laser melting of weak-textured commercially pure titanium with high strength and ductility: A study from laser power perspective

Li X.P., ..., Kruth J.P.

Materials and Design • Volume 116 • 2017[Show abstract](#) 

Research article

A study of keyhole geometry in laser welding of zinc-coated and uncoated steels using a coaxial observation method

Kim J., ..., Ki H.

Journal of Materials Processing Technology • Volume 225 • 2015 • Article 14466[Show abstract](#) [View all citing articles on Scopus](#) 

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